

AVNEET SINGH

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Summary

Computer Science Engineer dedicated to building **performant, AI-augmented web applications**. Expert in **Full-Stack development with a Frontend-first approach**, specializing in **React, JS/TS, and RAG-integrated interfaces**. Passionate about clean architecture, responsive design, and transforming complex engineering challenges into elegant user experiences.

Experience

The University of Toledo: College of Engineering

October 2023 - Present

IT System Administrator

Toledo, Ohio

- **Secured Windows and Linux environments** by managing **Active Directory** group policies and user permissions to support high-performance computational workloads.
- **Strengthened hardware asset security** by implementing TPM 2.0 protocols and **PowerShell-based** automated tracking, effectively reducing overhead costs by \$20,000 while hardening endpoint integrity against unauthorized access.

KPIT Technologies Limited

June 2025 - August 2025

Systems Engineer Intern

Novi, Michigan

- Developed real-time data-acquisition interfaces to bridge system metrics with monitoring tools, focusing on optimizing data flow and ensuring low-latency updates for status dashboards.
- **Engineered a modular middleware library** using Git-based version control for collaborative development, implementing robust error-handling and reusable logic to streamline data between complex subsystems.

KPIT Technologies Limited

June 2024 - August 2024

Software Engineer Intern

Novi, Michigan

- Designed an interactive **React** dashboards using component-driven UI logic to visualize complex ECU state data, implementing real-time updates and user-driven filtering that reduced manual reporting time by 70%.
- Engineered cross-platform visualization workflows aligned with OEM standards, leveraging modular JavaScript logic to synchronize responsive web interfaces (React) with desktop diagnostic tools (PyQt5).

Softage Information Technology Limited

June 2023 - July 2023

Web Developer Intern

Gurugram, India

- Developed a responsive full-stack web application using Django, HTML, CSS, JavaScript, and Python, managing both frontend and backend functionality.

Projects

RAG OpenPDF AI ChatBot | *FastAPI, React, LangChain, OpenAI, FAISS*

August 2025

- **Architected a responsive React interface** for real-time AI interaction, implementing streaming text responses and complex state management to handle unstructured PDF data queries via a RAG pipeline.
- **Optimized the frontend data-fetch layer** to synchronize with FAISS vector searches, ensuring low-latency rendering of context-relevant results and a seamless user experience for document analysis.

Moodify – AI Mood-Based Music Player | *React.js, FastAPI, PostgreSQL, Node.js, OpenCV*

May 2025

- Engineered a **high-performance React (Vite) interface** that integrates real-time facial emotion classification, managing complex state transitions based on dynamic Computer Vision triggers.
- **Optimized the frontend-to-model inference pipeline**, ensuring low-latency UI updates and seamless media playback synchronization across diverse emotional state changes.

Network Report Generation Tool | *Python, Pandas, Numpy, PyQt5*

August 2024

- **Engineered a high-performance data visualization suite** using PyQt5 to automate complex network log analysis, leveraging vectorized transformations to reduce manual reporting time by 60%.

Education

The University of Toledo: College of Engineering

Anticipated Graduation Date: May 2026

Bachelor of Science: Computer Science and Engineering

Dean's List: Spring 2023, Spring 2024

Skills

Frontend Development: React (Vite, Hooks), JavaScript, TypeScript, HTML, CSS3, Tailwind CSS, PyQt5, UI/UX

AI & Machine Learning: RAG Architecture, LangChain, OpenAI API, FAISS, OpenCV, Computer Vision

Languages: JavaScript, Python, Java, C, SQL, Lua, HTML/CSS

Backend & Data: FastAPI, Node.js, Django, RESTful APIs, Python (Pandas, NumPy), Vectorized Data Processing, SQL

Tools & Cloud: Git, GitHub, Docker, Azure, VS Code, PowerShell, MySQL, MongoDB, SSMS

Soft Skills: Technical Leadership, Collaborative Problem-solving, Cross-functional Communication, Meticulous Engineering